NOT MEASUREMENT SENSITIVE

O-S-635E
20 September 2007
SUPERSEDING
O-S-635D
4 May 1977

FEDERAL SPECIFICATION

SODIUM POLYPHOSPHATES, TECHNICAL

Inactive for new design after 29 November 2001.

The General Services Administration has authorized the use of this federal specification by all federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers three types and three forms of sodium polyphosphates (see 6.1).

1.2 Classification. Sodium polyphosphates shall be of the following types and forms.

1.2.1 Type. The type of sodium polyphosphates shall be as specified (see 6.2(b)).

Type I - Sodium tetraphosphate
Type II - Sodium hexametaphosphate
Type III - Sodium tripolyphosphate

1.2.2 Forms. The form of the sodium polyphosphates shall be as specified (see 6.2(c)).

Form 1 - Powder
Form 2 - Granular (crushed)
Form 3 - Plate

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: STDZNMGT@dla.mil or Defense Supply Center Richmond, ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616.

AMSC N/A FSC 6810

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2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standards

FED-STD-313 - Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities.

(Activities outside the federal government may obtain copies of federal specifications, standards, and commercial item descriptions as specified in the General Information section of the Index of Federal Specifications, Standards and Commercial Item Descriptions. The index is for sale on a subscription basis from the General Services Administration, Federal Supply Service, Specification Section, East 470 L'Enfant Plaza SW, Suite 8100, Washington, DC 20407.)

(Single copies of this specification, and other federal specifications and commercial item descriptions required by activities outside the federal government for bidding purposes are available without charge from the General Services Administration, Federal Supply Service, Specification Section, East 470 L'Enfant Plaza SW, Suite 8100, Washington, DC 20407.)

(Federal government activities may obtain copies of federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards

MIL-STD-2073-1 - Standard Practice for Military Packaging.

(Copies of military specifications and standards required by contractors in connection with specific procurement functions are obtained from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094. Electronic copies of specifications and standards may be obtained from http://assist.daps.dla.mil.)

Code of Federal Regulations (CFR)

29 CFR 1910.1200 - Hazard communication.

(The Code of Federal Regulations (CFR) is for sale on a subscription basis from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. When indicated, reprints of certain regulations may be obtained from the federal agency responsible for issuing them. Electronic copies may be obtained from http://www.access.gpo.gov/.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

ASTM International

ASTM D 501	- Standard Test Methods of Sampling and Chemical Analysis of Alkaline Detergents.
ASTM D 502	- Standard Test Method for Particle Size of Soaps and Other Detergents.
ASTM E 11	- Standard Specification for Wire Cloth and Sieves for Testing Purposes.
ASTM E 70	- Standard Test Method for pH of Aqueous Solutions With the Glass Electrode.

(Private sector and civil agencies may purchase copies of these voluntary standards from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959. Electronic copies may be obtained from http://www.astm.org/.)

3. REQUIREMENTS

3.1 Particle size.

3.1.1 Forms 1 and 2. Forms 1 and 2 shall conform to the particle size as specified in table I when tested as specified in 4.4.3.1.1.

TABLE I. Particle sizes of form 1 and form 2.

Sieve size	Percent retained on sieve	
	Form 1	Form 2
3.36 mm (no. 6) sieve	--	10 (max.)
600 µm (no. 30) sieve	5 (max.)	--
150 µm (no. 100) sieve	40 (max.)	80 (min.)

3.1.2 Form 3. Form 3 may be irregularly shaped plates and shall measure not greater than 4 inches (10.2 centimeters (cm)) on an edge and 1/4 inch (0.6 cm) in thickness when tested as specified in 4.4.3.1.2.

3.2 Chemical requirements. Sodium polyphosphate shall conform to the requirements of table II when tested as specified in 4.4.3.2 through 4.4.3.4.

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TABLE II. Chemical requirements for sodium polyphosphates.

Requirement	Type		
	I	II	III
Total P_2O_5, weight percent, minimum	62.8	66.5	56.0
pH, 1 percent solution	7.2 - 8.3	6.6 - 7.3	9.3 - 10.4
Matter insoluble in water, weight percent, maximum	0.1	0.1	0.1

3.3 Workmanship. Sodium polyphosphate shall be visibly free from foreign matter and suitable for its intended use when examined as specified in 4.3.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the acquisition order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the acquisition order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Quality conformance inspection shall consist of the examinations of paragraph 4.3 and the tests of 4.4.

4.3 Examination.

4.3.1 Examination of preparation for delivery.

4.3.1.1 Lot. For this examination, a lot shall be sodium polyphosphate fully packed, marked, and offered for examination at one time. Lot size shall be expressed in number of filled shipping containers.

4.3.1.2 Sampling plan. Sample unit for this examination shall be one shipping container. Defects shall be scored in accordance with table III.

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Table III. Examination of preparation for delivery.

Examine	Defect	Result
Marking	Missing, incorrect, illegible, not as specified.	Any defect shall be cause for rejection of the entire lot.
Packing	Not as specified	
Contents	Net contents less than specified	
Workmanship	Container bulged, distorted, or otherwise defective; tear out or hole; evidence of leakage from any part of the container	

4.3.2 Examination of workmanship.

4.3.2.1 Lot. For this examination, a lot shall be one type and form of sodium polyphosphate produced by one manufacturer, at one plant, from the same raw materials, by the same manufacturing process, and offered for examination at one time. Lot size shall be expressed in the number of pounds of sodium polyphosphate.

4.3.2.2 Sampling plan. Sample size for this examination shall be one pound (approximately 500 grams) drawn at random from the lot. The lot shall be rejected if the sample contains foreign matter or is not suitable for its intended use.

4.4 Testing.

4.4.1 Lot. A lot shall be as specified in 4.3.2.1.

4.4.2 Sampling plan. Sampling shall be as specified in 4.3.2.2. The sample shall be sealed in a clean, dry, moisture-proof container with identifying lot number and date. The sample shall be tested as specified herein. The lot shall be rejected if the sample fails one test.

4.4.3 Test methods.

4.4.3.1 Particle size.

4.4.3.1.1 Forms 1 and 2. Particle size of forms 1 and 2 shall be determined in accordance with ASTM D 502 except that sampling shall be as specified in 4.3.2.2 and sieves shall be as specified in table I. Sieves shall conform to ASTM E 11. Percentage of sodium polyphosphate retained on the sieves shall be calculated by dividing the weight of the residue on the sieve by the original weight of the sample and multiplying by 100.

4.4.3.1.2 Form 3. Particle size of form 3 may be visually estimated.

4.4.3.2 Phosphorus. Total phosphorus, as P_2O_5, shall be determined in accordance with ASTM D 501 for sodium triphosphate, total P_2O_5 by pH titration.

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4.4.3.3 pH. The pH of a one percent solution shall be determined in accordance with ASTM E 70.

4.4.3.4 Insolubles. The insoluble matter shall be determined in accordance with ASTM D 501 for sodium triphosphate, matter insoluble in water.

5. PACKAGING

5.1 Preservation, packing, and marking. The sodium polyphosphate shall be preserved, packed, and marked in 50 pound (lb) or 10 lb units in accordance with MIL-STD-2073-1 as stated in the solicitation, contract, or purchase order (see 6.2(d)).

5.2 Precautionary marking. Each unit container shall bear a label printed in bold type and securely affixed to the unit container as follows:

NOT FOR USE IN POTABLE WATER

6. NOTES

INFORMATION FOR GUIDANCE ONLY. (This section contains information of a general or explanatory nature that is helpful, but is not mandatory.)

6.1 Intended use. Sodium polyphosphates covered by this specification are intended for use in industrial water treatment but are not intended for use in water for human consumption. They form soluble complexes with calcium and magnesium and may be used to soften water and control scale formation in condensers, heat exchangers, pipelines, and boilers. Sodium polyphosphates are interchangeable for the intended use but possess the differences indicated below. Selection of the type and form should depend upon established patterns of usage, compatibility with equipment, and economic advantage.

6.1.1 Type I. Sodium tetraphosphate is primarily intended for use in water treatment for laundries.

6.1.2 Type II. Sodium hexametaphosphate is primarily intended for use in the treatment of boiler feedwater. This type is commonly used for the threshold treatment of industrial water. In those cases where the use of this type requires the addition of a relatively large quantity of caustic soda or soda ash, type III might be preferred.

6.1.3 Type III. Sodium tripolyphosphate is intended for use as a substitute for types I or II whenever it is determined to be necessary for performance or economically advantageous.

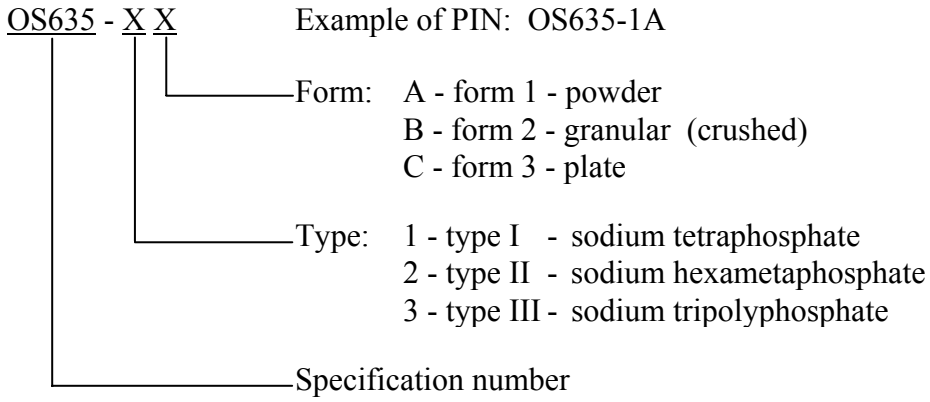
6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, revision, and date of this standard.
b. Type required (see 1.2.1).
c. Form required (see 1.2.2).
d. Preservation, packing, and marking requirements, (see 5.1).

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6.3 Materials safety data sheet (MSDS). Contracting officers will identify those activities requiring copies of completed MSDS prepared in accordance with FED-STD-313 and meeting the requirements of 29 CFR 1910.1200. The pertinent government mailing addresses for submission of the data are listed in FED-STD-313 and 29 CFR 1910.1200 requires that the MSDS for each hazardous chemical used in an operation must be readily available to personnel using the material. Contracting officers will identify the activities requiring copies of the MSDS.

6.4 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



6.5 Subject term (key word) listing.

boiler feedwater
hexametaphosphate
laundries
tetraphosphate
tripolyphosphate
water treatment

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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MILITARY INTERESTS:

Custodians:
Army - MD1
Navy - SH
Air Force - 68

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - FSS

Preparing activity:
DLA - GS3

(Project 6810-2007-001)

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